

Bryson Owens

Virginia Commonwealth University
Department of Mathematics and Applied Mathematics
Email: owensb7@vcu.edu
Website: owensbc.github.io

Education

- **August, 2025 – Present:** Ph.D. Student, Virginia Commonwealth University
- **August, 2024:** M.S. Mathematics, University of Illinois at Chicago
- **May, 2020:** B.S. Mathematics, Minor in Spanish, Colorado State University

Research

- **Research Interests:** My current research interests are in enumerative algebraic geometry; specifically, in intersection theory on moduli spaces of curves.
- **Fall 2025 – Present:** Research Project – Infinite-Dimensional Lie Algebroids and Moduli of Principally-Polarized Abelian Varieties
- **Spring 2020 – Fall 2020:** Research Project – Intersection Theory on Moduli Spaces of Curves, generalizing results from Summer 2019, advised by Dr. Renzo Cavalieri
- **Summer 2019:** REU at Colorado State University – Intersection Theory on Moduli Spaces of Admissible Covers, advised by Dr. Renzo Cavalieri

Publications

1. *Boundary Expression for Chern Classes of the Hodge Bundle on Spaces of Cyclic Covers*, with Seamus Somerstep and Renzo Cavalieri. *Involve, a Journal of Mathematics* 14-4 (2021), 571–594. DOI 10.2140/involve.2021.14.571

Accepted for Publication

1. *All the λ_1 's on Cyclic Admissible Covers*, with Seamus Somerstep and Renzo Cavalieri. To appear in *Proceedings of the AMS*.

Teaching

- **Fall 2025:** Teaching Assistant – MATH 131, Contemporary Mathematics, VCU
- **Summer 2025:** Instructor – Summer Enrichment Math Workshop, UIC
- **Spring 2025:** Teaching Assistant – MATH 110 College Algebra, UIC
- **Fall 2024:**

- Instructor of Record – MATH 182 Emerging Scholars Workshop for Calculus II, UIC
- Teaching Assistant – MATH 181 Calculus II, UIC
- **Summer 2024:** Instructor – Summer Enrichment Math Workshop, UIC
- **Spring 2024:** Teaching Assistant – MATH 181 Calculus II, UIC
- **Fall 2023:** Instructor of Record – MATH 090 Intermediate Algebra, UIC
- **Summer 2023:** Teaching Assistant – MATH 180 Calculus I Summer Edge Program, UIC
- **Spring 2023:**
 - Instructor of Record – MATH 182 Emerging Scholars Workshop for Calculus II, UIC
 - Teaching Assistant – MATH 181 Calculus II, UIC
- **Fall 2022:** Teaching Assistant – MATH 121 Precalculus, UIC
- **Fall 2021:** Teaching Assistant – MATH 180 Calculus I, UIC

Talks and Presentations

- **September 2025:** Richmond Geometry Seminar Poster Session – *Chern Classes of the Hodge Bundle on Spaces of Admissible Covers*
- **September 2025:** Virginia Commonwealth University Student Geometry Seminar – *Examples of Infinite-Dimensional Lie Algebras*
- **March 2025:** Virginia Commonwealth University Geometry Seminar – *Chern Classes of the Hodge Bundle on Spaces of Cyclic Admissible Covers*
- **September 2024:** UIC Minicourse on Syzygies – *Castelnuovo-Mumford Regularity*
- **January 2024:** UIC Commutative and Homological Algebra Graduate Seminar – *Structure of Smooth, Unramified, and Étale Homomorphisms*
- **March 2023:** UIC Commutative and Homological Algebra Graduate Seminar – *Cohen-Macaulay Rings and Flatness*
- **February 2023:** UIC Commutative and Homological Algebra Graduate Seminar – *Introduction to Local Cohomology*
- **October 2021:** UIC Graduate Algebraic Geometry Seminar – *Intersection Theory and Admissible Covers*
- **January 2020:** JMM Undergraduate Poster Session – λ_1 on Spaces of Admissible Covers

Grants

- **Fall 2025 – Present:** NSF Award DMS-2404896
- **Spring 2022 – Summer 2022:** NSF RTG Pre-doctoral Fellowship – UIC
- **Summer 2019:** Research Grant – CSU